

Addressing data scarcity in speech emotion recognition: A comprehensive review

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ABSTRACT

Speech emotion recognition (SER) is a critical field within affective computing, aiming to detect and classify emotional states from speech signals, which vary dynamically over time. These signals encode complex relationships between features at multiple time scales, effectively reflecting a speaker's emotional state. Despite significant progress, SER faces the persistent challenge of labeled data scarcity, a major obstacle given the data-intensive requirements of deep learning models. This scarcity often results in small, imbalanced datasets that hinder model generalization. Various strategies, including feature selection, data augmentation, domain adaptation, and fusion techniques, have been employed to mitigate these issues. However, comprehensive reviews that critically analyze these methods remain limited. In this paper, we provide an extensive review of these data scarcity strategies in SER, assessing their merits and limitations in terms of efficiency and robustness. Special attention is given to how these strategies enhance the performance of both acoustic and multimodal SER systems when operating on limited datasets. Additionally, we highlight the potential of fusion strategies combined with attention mechanisms as promising solutions to improve convergence and reduce model complexity.

1. Introduction

Artificial intelligence (AI) techniques have been applied in most aspects of daily life to enhance human interaction with technology. AI has been incorporated into behavior recognition to assist human beings in several aspects of life like support, sentiment analysis, and reviews. Some of the current examples that attempt to act upon the users' intent are chatbots, virtual private assistants, and virtual agents. Since 1997 when Prof. Picard introduced affective computing [1], advancement in research to enhance emotion recognition systems has been in high gear. This has been in terms of recognition, synthesis, and enhancement of emotions [2]. Emotion recognition is applied in education, customer sentiment analysis [3], health monitoring especially mental health, and patient mood monitoring [4,5] among others. Emotions can be classified according to discrete categories or continuous emotional dimensions. Ekman and Friesen [6] categorized the discrete emotions as happiness, sadness, surprise, anger, disgust, and fear. According to Tsiourti et al. [7], the discrete categories of emotions can also be categorized in terms of a two-dimensional plane of valence and arousal. Verma et al. [8] further proposed to include dominance and analyze the emotions in a three-dimensional continuous space.

To improve the real-time performance of Speech emotion recognition (SER) systems, researchers have proposed supervised, unsupervised, and semi-supervised approaches for learning salient representations of multi-modality data. This is possible due to the growing availability of data in the form of images, speech signals, text, and physiological signals. These approaches are important for the model to learn not only the homogeneous feature of diverse data but also the heterogeneity that exists between them to enhance the performance. However, this is only possible with the availability of large datasets since it enhances the ability of the model to learn diverse patterns and generalize better on unseen data. Large emotional datasets that encompass a variety of attributes like speaking style, race, age, and gender provide rich patterns to the deep learning models.

The scarcity of emotional speech data is an obstacle to the progress of SER research and needs to be alleviated. With deep learning approaches, inadequate training data yields poor performance while large datasets can ensure robust outcomes in both lab experiments and real life. Those models that perform well during laboratory experiments often lack generalization across different environments and conditions. Therefore, the data scarcity problem hampers the application of deep

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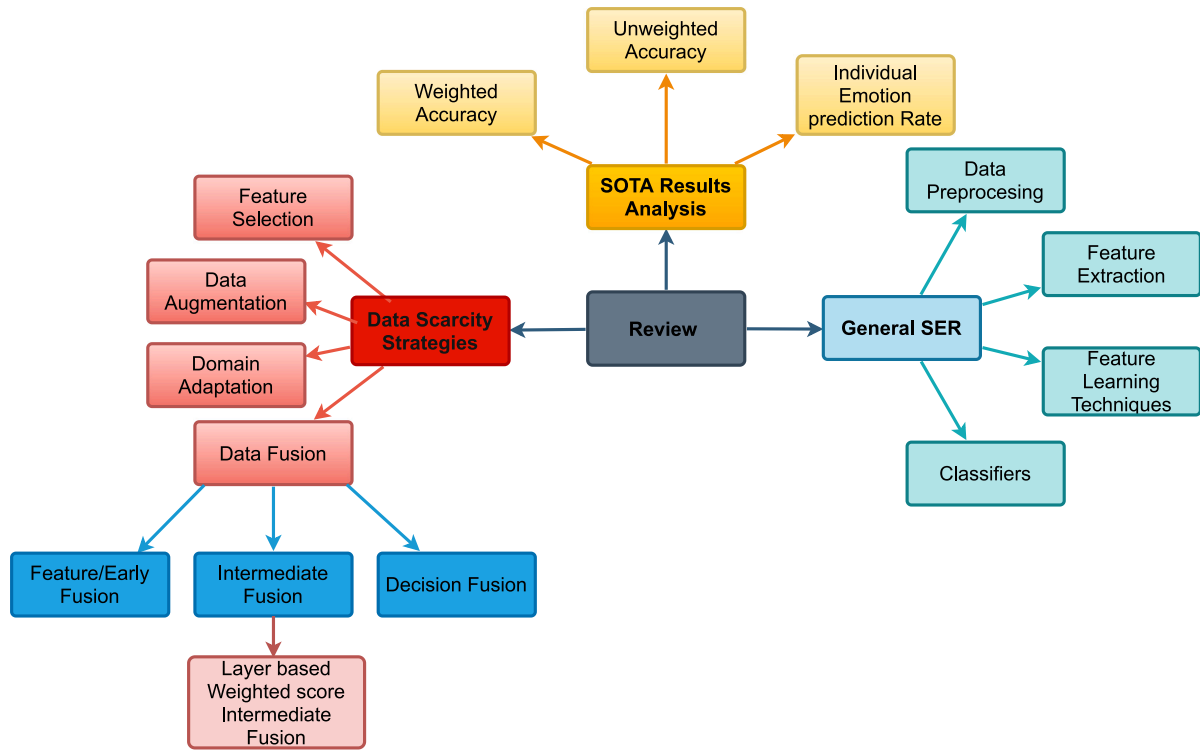


Fig. 1. The structure of the review.

learning techniques in SER since their performance is low and encounters overfitting. Small, imbalanced datasets and lack of generalization by trained SER models are some of the data scarcity challenges that need to be solved to improve the performance of SER systems.

Bearing in mind the fact that large datasets enable models to learn and identify diverse patterns, they would be the ultimate solution for SER data scarcity problems. However, the unavailability of emotional speech data for some complicated emotion states like disgust and calmness coupled with the difficulty of data collection for most speech emotion states that is costly and expertise intensive during annotation makes it hard. The small and imbalanced datasets challenge caused by data scarcity is evident in the publicly available limited datasets that researchers use to evaluate the new approaches they usually propose. The German language emotional (EMODB) dataset [9] consists of 493 utterances, and the American English language Ryerson audio-visual database of emotional speech and song (RAVDESS) dataset [10] consists of 1,440 files for speech and 1,012 for songs. The British English language Surrey audio-visual expressed emotion (SAVEE) datasets [11] contain 480 utterances. The other datasets that have slightly more than 10,000 utterances are the interactive emotional dyadic motion capture (IEMOCAP) [12], multimodal emotion lines dataset (MELD) [13], and the CMU multimodal opinion sentiment and emotion intensity (CMU-MOSEI) [14] datasets. The Toronto emotional speech set (TESS) datasets [15] which consists of balanced datasets with 200 samples per emotion have short phrases which may instead affect the generalization of models on longer utterances. The articles that we reviewed in this paper also use the Malayalam, crowd sourced emotional multimodal actors (CREMA), audio, speech, and vision processing lab emotional sound (ASVP-ESD), the Italian emotional speech (EMOVO) and the emotional speech (ESD) datasets. Though review articles about emotion recognition exist [16], those that particularly review data scarcity strategies especially for speech are rare. This motivated us to review the latest solutions used in SER systems to solve the data scarcity problem. We subsequently endorse fusion strategies as the ultimate solution for deep learning-based SER systems.

Over time, researchers have proposed different ways of dealing with data scarcity in deep learning-based SER applications that include

feature selection, domain adaptation, data augmentation, and fusion strategies. In terms of feature selection, unsupervised methods that include principal component analysis (PCA), feature clustering, spectral feature selection, independent component analysis (ICA), evolutionary algorithms, filter methods, and attention mechanisms have been used to solve the data scarcity challenge.

Unsupervised methods learn emotional speech representations from the available unlabeled speech that is often collected with no emotional detection intentions. Variational encoders, adversarial autoencoders, adversarial variational Bayes, and denoising autoencoders were investigated in [17] and they found that SER systems benefit from variational methods in learning features as a solution to data scarcity challenges. In terms of domain adaptation, Huijuan et al. [18] solved the challenge of reduction of transfer efficiency caused by class distribution discrepancies by proposing a local domain adaptation approach where they use a category-grained discrepancy for distance computation between the source and target domains. In [19], transfer learning based on time delay neural networks (TDNN) was proposed for SER in Malayalam, a less resource language. It is premised on the fact that people may share similar characteristics in their vocal output. We, however, opine that this fact depends on the culture, age and race of the person. Besides, even the presumed resource languages have datasets with fewer samples, especially for non-neutral emotions. Semi-supervised methods have also been used as solutions to SER data scarcity challenges. In [20], a semi-supervised model that extracts quaternion embedding from spectrograms to train an SER system that consists of an encoder, classifier, and decoder was proposed for emotion detection. Another semi-supervised method is a multi-task approach which involves learning shared representations as a solution for data scarcity. Such an approach was proposed in [21] by allowing the model to learn generalized representations through the use of augmented data and unsupervised reconstruction. The fusion of either features in acoustic SER or different modalities in multi-modality SER has also been used to solve the data scarcity problem. In [22], features were fused at an early stage to maximize them for commendable performance. In [23,24], transformers were deployed at different stages of learning

to optimize the acoustic and text features from the small datasets called IEMOCAP for commendable performance. However, it is worth noting that a careful consideration of the relationships between different modalities of data helps to avoid negative transfer and understanding of knowledge [25].

Though advantageous, the mentioned strategies have limitations that include but are not limited to the need for handcrafted features, large amounts of data, and domain mismatch. The change of emotional states due to augmentation, emotional complexity, and ambiguity escalates the problem of generalization further. Some researchers have attempted to solve these by combining two approaches. Gao et al. [26] utilized autoencoders and feature-level fusion to enhance better performance of SER systems. In the bid to analyze the data scarcity challenges and existing solutions in both acoustic and bimodal SER systems, we answer the following question in this paper.

- What are the various strategies deployed to solve data scarcity in deep learning-based SER systems?
- How effective are these strategies in addressing the data scarcity challenge of SER systems?
- How have the strategies performed in situations where they have been deployed in literature?
- Which strategy is the most effective in addressing the data scarcity problem in deep learning-based SER systems?

As shown in Fig. 1, the rationale and structure of our review include a review of general speech emotion recognition (GSER) concerning limited datasets, data scarcity strategies utilized in literature, and the state-of-the-art results analysis and discussion. However, we dwell more on the data scarcity strategies and the analysis of their results. We refer the reader to a detailed general SER review presented in [27]. The research works reviewed in this paper were majorly searched from Google Scholar, IEEEExplore, Science Direct, and nature repositories. The following keywords were used during the search for relevant articles; limited datasets, data scarcity, speech emotion recognition, and reviews on data scarcity. A total of 200 relevant articles were collected. These articles were screened to sort the essential and most relevant articles to our goal. The subject, goal, year of publication, and relevancy to the four major data scarcity strategies were used for screening. We mostly reviewed the relevant articles published in the past ten years. They were divided into the following four major categories; GSER, feature selection, data augmentation, domain adaptation/transfer learning, and fusion strategies. As shown in Fig. 2, 110 out of 150 articles were chosen after screening. This represents 73.3% of the downloaded articles at the start of the review. These included 30.9% for general SER, 11.8% for feature selection, 13.6% for data augmentation, 7.3% for domain adaptation/transfer learning, and 36.4% for fusion strategies. The high number of fusion strategies shows that it is the most common strategy utilized in the literature however, recently it has been used in combination with other strategies.

We not only review the multi-modal SER data fusion but we also analyze the impact of unimodal unique feature fusion which to the best of our knowledge has not been reviewed before. We endorse the use of fusion strategies and co-learning strategies for both heterogeneity features within the acoustic modality and multi-modality scenarios for SER. We also introduce a different perspective of intermediate fusion where the shallow, intermediate, and higher-level features are fused in what we term as layer-based weighted score fusion. We opine that fusion strategies of the feature and modality representations are important for SER systems under data scarcity scenarios.

Summarily the contributions of this paper are threefold;

- A comprehensive review of feature selection, data augmentation, domain adaptation, and data fusion as data scarcity strategies for SER is presented.

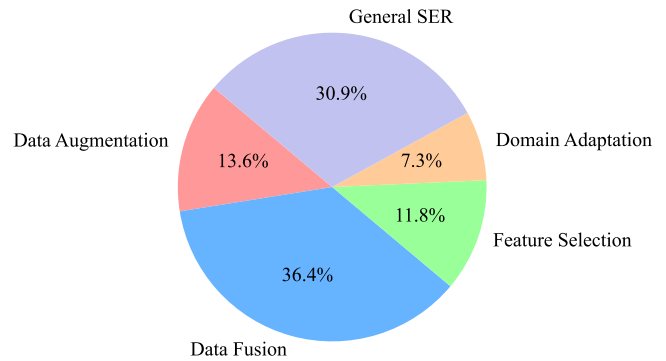


Fig. 2. The proportion of articles reviewed for the different SER data scarcity strategies.

- We carry out a performance review analysis of the different data scarcity strategies proposed in the literature discussing the merits and demerits of each strategy in terms of efficiency and robustness.
- We also endorse layer-based fusion approaches for both acoustic heterogeneity features and multi-modality fusion for SER systems.

This makes our article unique from the few related articles about deep learning-based strategies for speech emotion data scarcity. To the best of our knowledge, this is the first comprehensive review of data scarcity strategies in the SER system.

The rest of the paper is organized as follows: a review of strategies for data scarcity in SER systems including feature selection, data augmentation, domain adaptation, and data/feature fusion are discussed in Section 2. We analyze and discuss the state-of-the-art (SOTA) results in terms of their significance to SER efficiency and robustness in Section 3. The paper is concluded in Section 4.

2. Strategies for data scarcity ser systems

In this section, we review strategies used in state-of-the-art research works that provide solutions to the challenges of SER data scarcity. The major challenges include small datasets availability, imbalanced datasets, and lack of generalization that affects the model's performance. In recent works, these strategies have been used singly or in combination as we discuss in this section.

2.1. Feature selection in SER systems

Feature selection in deep learning models is an essential process to enhance model performance, reduce overfitting, and decrease training time thereby solving the data scarcity challenges. Although deep learning models can inherently learn important features, feature selection techniques can still be applied to improve model efficiency and interpretability since salient and emotionally rich features are needed for better performance in data scarcity situations. Feature selection can also be considered as an unsupervised method for selecting relevant features from SER datasets without label considerations due to data scarcity. Generally, feature selection methods include embedded methods, filter methods, wrapper methods, dimensionality reduction methods, attention mechanisms, and feature importance methods.

The earlier work in this field used PCA on a variety of combined sound features to form a selected feature subset. Its usage can be observed in [28,29], and [30]. With promising results, the work in [31, 32] combined PCA and neural networks where the PCA was used for feature parameter reduction and redundancy elimination while the neural network was the classifier. Though PCA is good at the identification of uncorrelated features, dimensionality, and noise reduction, it has some drawbacks. PCA may lead to loss of information since it only retains components with the highest variance. In addition, the selected

principal components are linear combinations of original features and may not be easily interpretable and may not capture complex nonlinear relationships in the SER data.

Correlation feature selection (CFS), is an alternative feature selection method that was utilized in [33,34]. In the work of [33], CFS is applied after extracting deep features using a deep convolutional neural network (DCNN). CFS method selects discriminative features out of a subset of attributes that have high correlation with the class labels for emotional detection. The work of [34] involved extraction of 64 statistical speech features and the use of mean impact value and CFS for relevant feature subset selection.

Other researchers used filter feature selection methods which evaluate the importance of features based on the intrinsic properties of the data. Haider et al. [35] describes the use of infinite latent feature selection (ILFS), ReliefF, and generalized Fisher score and their approach that they call active feature selection method. These methods are good in the reduction of the number of features which in turn benefits the memory and computational resource usage. In doing that, these feature selection methods utilize the most salient features significant for emotional detection even with small datasets like EMODB, SAVEE, and EMOVO. The results reported for all these datasets show that commendable performance can be achieved when a subset of the feature set selected using the above-mentioned methods is used compared to a full feature set.

Evolution algorithms have also been used for feature selection as a solution to data scarcity challenges in SER. The multi objective genetic algorithm reported in [36] utilizes the multi optimization method based on differential evolution for effective feature selection. In [37], an extreme learning machine (ELM) based on genetic algorithm is used. Mel frequency cepstral coefficients (MFCCs) are extracted and the ELM based on a genetic algorithm for salient feature selection is used for emotional classification. The earlier optimization work on ELM [38,39] ignored the optimization of random hidden weights and bias. However, the genetic algorithm-based ELM operates by tuning the weight and bias of hidden nodes using selection, crossover, and mutation. Unlike the other feature selection methods, genetic algorithms can be adapted to complex, non-linear, and high-dimensional feature spaces. Nonetheless, they need to evaluate multiple generations of candidate features, which makes them computationally expensive and needs parameter tuning for population size, mutation rate, and crossover rate. In [40], a particle swarm optimization (PSO) algorithm has also been deployed to obtain an optimized feature vector for emotional classification. The obtained feature subset is dimensionally reduced and unique allowing the classifier to produce commendable results with fast response and low memory usage. PSO is superior to genetic algorithms since it consists of a lower computational cost due to the absence of crossover and mutation operations.

With the advent of deep learning, attention mechanisms replaced the highly computational resource-intensive traditional feature selection methods explained so far. Attention mechanisms have been pivotal in data scarcity situations since their operation involves the selection of the most significant information. The most relevant parts of the input data are utilized when making predictions, allowing the model to capture dependencies and patterns more effectively which is good for data scarcity situations. Additive [41], multiplicative [42], and the transformer-based attention mechanisms [43] are the baseline attention mechanisms in deep learning. As stated in [44], they were first applied in SER to obtain the most significant speech features in [45,46]. They are often used in combination with traditional deep learning techniques like convolutional neural networks (CNNs) and recurrent neural networks (RNNs). Besides the capabilities of CNNs, it is important to use them in combination with attention mechanisms since the datasets are small, and the use of transformers directly could cause slow or difficult model convergence. RNNs like gated recurrent unit (GRU) and long short-term memory (LSTM) variants learn long-term dependencies. Transformer-based multi-head attention mechanisms solve the data

scarcity challenges in SER by the use of parallel and dynamic operations to obtain context vectors. It has been deployed in [22,47–49] for acoustic SER. It has also been utilized to avoid data scarcity challenges in bimodal SER [23,27], and [24] among others. Because of the scarcity of data for driver's SER in smart cockpits, multi-head self-attention was applied in [50,51] to overcome the small and imbalanced data scarcity challenge. Particularly, in [51], it was used to extract significant global and local features from a driver's voice before fusion to estimate his/her emotions and establish the safety and readiness to execute the activity.

Attention mechanisms are also often used in combination with data augmentation and fusion to solve the data scarcity problem. The work in [52] includes multi-head cross attention for the deep fusion of acoustic and text features for SER. The cross attention is aimed at learning the inter-modality relationship representations between text and audio modalities while the deep fusion is applied at the intermediate level to learn the salient features that jointly represent emotion clues in both of the modalities. An average 4.5% improvement was registered on relatively small IEMOCAP and MELD datasets. In addition, though it increases the computational complexity, sparse autoencoders can be combined with attention mechanisms to benefit from both labeled and unlabeled data and therefore solve the data scarcity challenges of SER systems. It was shown in [53] that this combination leads to a 4% and 5% improvement in SER systems for Mandarin and English languages respectively. In [54] feature selection was also utilized in combination with early and late fusion approaches that we discuss later in this article. The histogram of gradients (HOG), open smile, and wavelet packet decomposition features were used in a two-stage feature selection. The two-stage feature selection involved the use of random forest at the first stage and grey wolf optimization at the second stage of selection. It should also be noted that, whether the highly computational or deep learning-based methods are used for feature selection, their success depends on the features extracted for emotional recognition.

2.2. Data augmentation in SER systems

Data augmentation is another solution for data scarcity challenges in SER systems. This technique enhances the diversity and volume of training data, thereby creating more robust models and improving model performance with limited available labeled emotional speech datasets hence enhancing model generalization. Data augmentation is applied in both intra-corpora and cross-corpora SER research works for data variations. Particularly, tremendous results were reported in the works of [55] in which matched and mismatched SER was investigated. Some of the common transformations applied to speech data to effect augmentation in SER literature include; noise injection, pitch shifting, time stretching (speed perturbations), equalization, reverberation, spectrogram augmentation (SpecAugment), mixup, vocal tract length perturbation (VTLP). Noise injection involves adding noise to the audio signal to simulate real-world environments. This is usually referred to as additive white Gaussian noise (AWGN) since it has constant power spectral density and follows a normal distribution in terms of its amplitude. Pitch shifting involves changing the pitch of the audio signal creating variations in the training data without altering the linguistic content. Speed perturbations involve altering the speed of the voice signal without changing the pitch. Equalization includes modification of the frequency response of the voice to emphasize or de-emphasize certain frequency bands. Reverberation is important for simulating different acoustic environments, like rooms of various sizes. SpecAugment is an augmentation technique that involves time warping, frequency masking, and time masking of the spectrogram or mel spectrogram of speech data. Mixup combines two audio samples and their corresponding labels to create a new training sample. VTLP simulates variations in a speaker's vocal tract length, altering formant frequencies. Qu et al. [56] solves the lack of children's emotional speech data by generating child-like voice spectra from an adult's voice

spectrum based on vocal tract length normalization (VTLN) and VTLP with a commendable recognition performance.

However, data augmentation techniques of the nature described above tend to enhance the bias that exists in the original data. This is why good performance can be reported in some works with poor confusion ratios. The poor confusion ratio also comes about if the augmentation technique chosen changes in such a way that the voice signals from one emotion state or emotion plane change into another that is different from its label hence confusing the model. Therefore, care needs to be taken when choosing the augmentation technique to be used in SER works for genuine benefits. Generalized emotional speech representations can be learned through the use of data augmentation. Speed perturbations, SpecAugment, and Mix-up [21] can be applied alongside multi-task learning. However, as explained earlier more effort is needed to achieve even better results for individual emotion states since the model tends to have a high confusion ratio for the emotions in the same emotional plane. A copyPaste algorithm [57] that combines voice signals of one emotional state with the neutral state to increase the number of labeled data can also be utilized. The performance evaluation shows that this approach is better than the use of the AWGN in terms of accuracy and individual emotion-state confusion ratios.

Nonetheless, with the advent of deep learning, generative and adversarial data augmentation techniques have been used to enhance the diversity and volume of training data as a solution to the data scarcity challenge. Unsupervised learning methods, such as generative adversarial networks (GANs), adversarial autoencoder (AAE), and adversarial variational Bayes (AVB) can capture the intrinsic structure of the data distribution in the learned feature representation. As noted in [58], adversarial auto encoders have a minimax game operation that allows them to exhibit generative model characteristics which are known for learning and generating descriptive salient features of data. In the investigation carried out in [17], a commendable performance obtained shows the impact of adversarial methods in audio data augmentation. The use of adversarial networks through the generation of more training examples or management of the distribution of these emotion states can solve the imbalanced nature of emotion state data. The work in [59] regularizes the distribution of latent representation to smoothen the emotion state boundaries by use of multi-instance learning-based adversarial autoencoder for classification. Therefore, tasks whose data is in abundance can be used to improve SER which is the primary task with the adversarial training being a secondary task.

The significance of adversarial-based data augmentation in improving SER was reported in [60] where a GAN and an autoencoder were trained. The outputs of the autoencoder and the generator were used as input to the discriminator obtaining the synthetic samples from the decoder. This arrangement produces samples in the latent space and synthetic samples in the original space.

The work in [61] uses GANs or their variants and adversarial autoencoders to generate more data or feature vectors that share the common latent representations with the original data before the primary and auxiliary classification tasks. Though this approach can be categorized as both data augmentation and domain adaptation, it overcomes the gradient vanishing problem in typical GANs and produces emotion-rich augmented samples. However, it is argued that synthetic samples tend to follow an arbitrary distribution rather than the actual data distribution. This was solved by replacing the cross-entropy adversarial loss used in other approaches with the Wasserstein divergence loss as a solution to the distribution challenge. A GAN model called HiFi-GAN neural vocoder [62] and a pre-trained model called wav2vec 2.0 [63] were used in [64] to synthesize emotionally rich data and test their quality. The GAN model was used to synthesize speech data from the available emotional voices database (EmoV) dataset. The wav2vec 2.0 model was fine-tuned using emotionally converted synthetic data, original data, and both of them. Commendable performance for speaker-dependent scenarios on the IEMOCAP and RAVDESS datasets was reported. It should however be noted that the performance

in the speaker-independent scenarios needs extra effort. On the other hand, Selvi et al. [65] argues that recreating emotional speech with limited labeled data is inefficient. They therefore separately synthesize non-emotional (unlabeled) speech and voice emotions (labeled), combine them, and then pass them to the cycle GAN for the generation of emotional speech. While the former approach may yield efficient emotional speech, it is computationally expensive and needs a lot of examples compared to the latter. A cycleGAN is used in the work of [66] to adversarially learn two domain adaptation functions for a pair of emotions for example functions for neutral to angry and angry to neutral emotion states which are then used for data augmentation for SER. They report a GAN training loss improvement of 2.51 which further affirms the significance of GANs in realistic speech data generation.

Though GANs are very good at synthetic emotional speech generation as compared to traditional speech transformations, the evaluation of the quality of the generated emotional data samples still has room for improvement. In addition to the GAN-based generation of emotional speech data from a mixture of emotional samples, Sahu et al. [67] used different metrics to evaluate the quality of the synthetic voice data samples and their performance. They tested the quality of the synthetic data by comparing the accuracy of SER classifiers on synthetic and real data. In addition, they used the Parzen window density estimation (PWDE) and the Fréchet inception distance (FID) that are common in computer vision to evaluate generated voice data giving a commendable analysis of the real and generated voice data. The quality of generated speech can also be evaluated by extracting spectral, prosodic, and energy features and investigating their emotional richness by using emotional recognition classifiers. The work in [68] evaluated the quality of the generated speech by the CycleGAN and StraGANv2 voice conversion models using the Mel-frequency cepstral distortion (MCD) and root-mean-square error (RMSE) for spectrum and prosody.

GANs are instance-based domain adaptation techniques since they generate synthetic data for the target domain using data from the source domain. However, in this paper, we separated their discussion from the domain adaptation strategy of data scarcity in SER because of how common they are used for data augmentation in SER research work. We discuss domain adaptation as a data scarcity strategy for SER systems in the next subsection.

2.3. Domain adaptation in SER systems

Domain adaptation is a deep learning-based technique used to transfer knowledge from the source domain to the target domain where there is limited data. In the recent SER research works, domain adaptation has improved model performance with limited labeled data by leveraging larger, related datasets. Generally, the approaches of domain adaptation applied to SER include feature, instance, and model-based domain adaptation. Since we summarily discussed instance-based adaptation in terms of data augmentation in the previous subsection we shall concentrate on feature and model-based adaptation SER research works as data scarcity strategies.

Adversarial methods alongside SER classification tasks have also been utilized for multi-task learning to improve model generalization. They enable learning of common features of the training data for all the tasks involved. In [69] adversarial auto encoding was used as an auxiliary task to learn significant and discriminative features thereby solving the data scarcity problem for emotion recognition. This approach not only shows good performance on intra-corpus categorical SER but also cross-corpus categorical and dimensional SER.

The deep scale fusion network proposed in [24] applies domain adaptation for data scarcity by use of pre-trained embeddings from the bidirectional encoder representations from transformers (BERT) for text embeddings, hidden units BERT (HuBERT) model for audio representations, and the emphasized channel attention, propagation

and aggregation (ECAPA) for speaker identification features. A similar approach is applied in [70] where same and cross-language scenarios are investigated using a model that is composed of decision-level fusion, attention mechanisms, and pre-trained models (wav2vec2.0 for audio and RoBERTa for text). This approach of combining fusion strategies with transfer learning solves the data scarcity problem since pre-trained models that were trained for similar tasks on larger datasets were used.

Though domain adaptation has been successful in computer vision, more effort in emotional speech to make sure that the adapted representation is emotionally rich and correctly represents the characteristics of distinct emotions. This is because the data from the source and target domains are often different resulting in a mismatch. In the work of Gao et al. [26], this mismatch in addition to other data scarcity challenges was solved by the use of a domain adversarial autoencoder that achieves domain-independent representations that are emotionally rich. An improvement of 4.15% of unweighted accuracy compared to the non-adversarial autoencoder approach on IEMOCAP datasets was achieved. Feng et al. [71] used few shot learning that involved fine-tuning the parameters of the siamese model to obtain the relative distance between samples rather than pattern distributions of emotions.

2.4. Fusion strategies used in SER systems

Fusion in deep learning-based SER systems can be categorized into feature and data fusion. This is because the fusion of multi-features and/or multi-modal data has yielded better results in recent works. In this sub-section, we explain feature and data fusion and explore the different fusion strategies and perspectives used in solving the data scarcity challenge in SER.

2.4.1. Multi acoustic feature fusion in SER systems

To obtain robust and accurate performance for SER systems, we can extract many kinds of features from the audio signals. These can be in terms of the time domain, frequency domain, or time-frequency domain. Time domain features include the zero-crossing rate (ZCR), root mean square energy (RMSE), and amplitude envelope (AE). Some of the frequency domain features include spectral centroid, spectral spread, and band energy ratio (BER). Mel frequency cepstral coefficients (MFCCs), mel spectrograms, chromagrams, and linear predictive cepstral coefficients (LPCCs) are some of the time-frequency domain features. An in-depth explanation of each feature together with their purpose is explained in [27]. Each of these features has a specific characteristic of sound that they depict for successful emotion recognition. The emotional information that is expressed by each feature is unique. The most common way of combining emotional speech features is through concatenation and inputting them into a model for feature learning and eventual classification. Since this is fusion at an early stage, the model performance may be affected by the challenges of early fusion. Pohjalainen et al. [72] fused MFCCs, formants, and short-term energy to achieve emotion recognition. Over time through several published research, it has been proved that the use of a combination of them yields better results. A multi-acoustic feature fusion of sub-mel spectrograms with the use of ResNET model branches is proposed in [73]. A six percent score improvement was reported when their model was tested on the emotional speech database for voice conversion research (ESD) dataset and also registered a commendable performance for the IEMOCAP dataset. An attention-based multi-learning model for SER is presented in [22]. The authors used an early fusion of mel spectrograms, MFCCs, and chromagrams with a commendable performance. However, it was evident from the independent class prediction rate results that there was a negative impact of the fusion strategy which would not bring out all the emotionally rich cues that exist in the unique features due to the early fusion strategy. An attention based head fusion model that uses fused time axis and frequency axis of the MFCCs was proposed in [74] with commendable

performance. The reported commendable performance shows that fusion within homogeneous or heterogeneous features can yield a good SER system performance. A remarkable performance is reported in [75] in which there is a fusion of constant-Q transform (CQT) spectrograms and mel spectrograms with STFT. The only drawback with this method is the complexity that the method introduces in training. Recently, Li et al. [76] enhanced SER systems through multi-scale feature fusion in which squeeze and excitation (SE) modules are used to select and weigh multi-scale features before fusion with the original information to benefit from the versatility of the emotional sound cues present in the MFCCs. Acoustic features can also be fused at the early stage of training with pre-trained features as seen in [77] to solve the challenge of weak robustness of features and overfitting of the model. From the literature explained so far it is evident that feature fusion can either be homogeneous (when there is the fusion of representation from the same feature) or heterogeneous (when there is the fusion of emotion representation from different features). In this paper, we endorse fusion solutions for heterogeneous feature fusion by use of layer-based weighted fusion.

2.4.2. Multi modal data fusion in SER systems

Depending on the source of data for training the models, the emotion understanding systems can be categorized as either single-modality or multi-modality. The sources of the emotional data can be images, speech signals, text, and physiological signals. A combination of some or all of these forms as model input sources gives rise to multi-modal emotion recognition. It is important to combine modality-specific, unique, and shared features at all levels of training. Ullah et al. [78] showed that combining text and emoticons improves sentiment analysis which confirms that different perspectives of data benefits the model's performance. Since speech emotions can be expressed in the form of audio and text, emotionally rich speech should involve both acoustic and lexical emotion cues especially if context and semantics are required to enhance SER performance [23,79]. The success of the transformer-based models has improved the embedding representation of text to include context and semantics which helps to blend well with acoustic features for robust SER systems. Priyasad et al. [80] proposed a model that uses self-attention with evaluation before and after intermediate-level fusion. The self-attention before fusion yielded a 5.2% weight accuracy improvement. Similarly, multi-head attention is used in [81] to fuse BERT features which are text representations, and MFCCs which are acoustic features at two levels of learning with commendable reported results compared to single modalities. Weighted late fusion of text and audio was proposed to learn complementary effects in [82] to enhance SER. However, there is insufficient multi-modal emotional information thereby causing less optimal interaction between modalities during fusion and training. An attempt to solve this challenge was made by Zhao et al. [24] through a deep-scale fusion network that uses pre-trained models. Their model also consists of a mutual transformer that is used to learn the shared multi-modality features for better emotion recognition. Other multi-data fusion approaches for limited SER data that our readers can interest themselves in include [47,83–85], and [86].

Multi-modal or multi-acoustic feature fusion allows the models to mimic human beings in terms of recalling and understanding emotions while engaging multiple senses during and after the learning process [87]. It should be noted that the world was modeled in multiple domains and therefore to understand its aspects one needs multiple perspectives or domains. In addition, emotions are subject-specific and can be influenced by language, culture, and the environment. This necessitates the need to have different perspectives of the emotional cues through different modalities obtained from a person's speech.

Nonetheless, SER data and acoustic feature fusion have several challenges that need to be carefully considered. These include robust representation, translation, alignment, and data sparsity. To solve these, the models should learn the homogeneous and heterogeneous

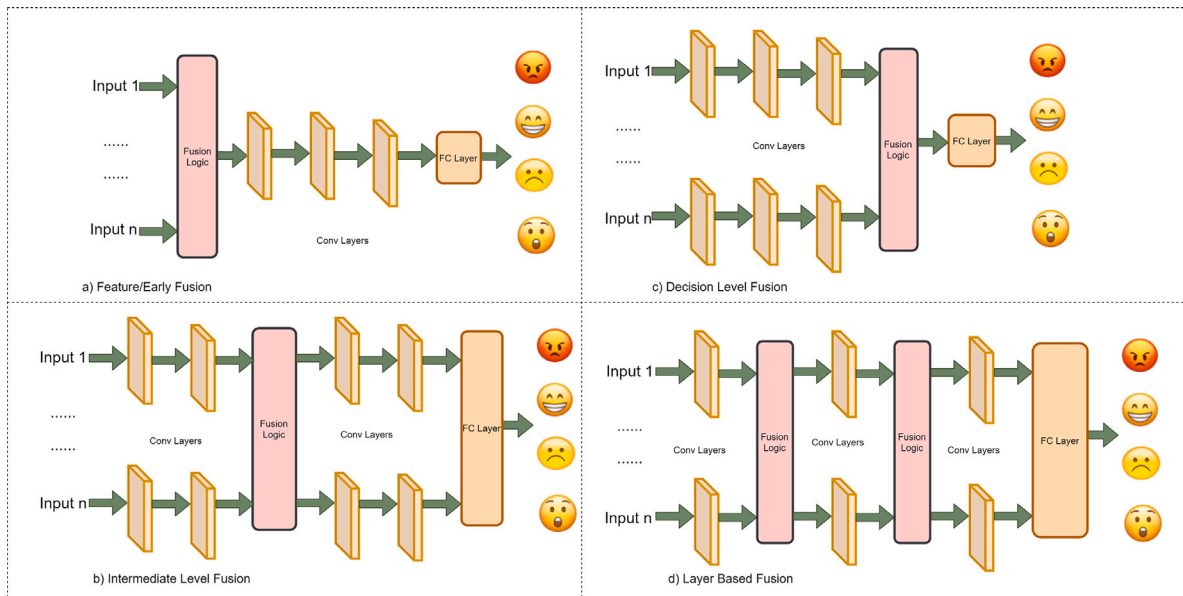


Fig. 3. The different fusion strategies commonly used in (SER) systems.

relationships of emotion states in each kind of features that exist in the different modalities. There exist heterogeneity and homogeneity features at the shallow, intermediate, and higher levels of the SER deep networks that need to be fused for better representation and eventual learning.

In the early days of multi feature/data fusion, early and late fusion were used in deep learning models. However, these two methods have many disadvantages that deprive the model of an opportunity to learn more salient features that benefit its performance in data scarcity. We explore the merits and demerits of the different ways to find marginal and joint representations of heterogeneous acoustic features and multi-modalities in a way that enables co-learning for better performance of speech emotion recognition systems with limited data. We analyze feature level, intermediate, and decision level fusion for speech emotion recognition for both multi-features and multi-modalities and endorse a solution for their drawbacks called layer-based weighted fusion.

As noted in [27], careful considerations are paramount for the fusion of different modality emotional information to ensure effective combination since they are usually implicitly aligned. This scenario is true for extracted heterogeneity acoustic features of the same modality too. Depending on the stage of fusion, the traditional fusion strategies in SER systems can be categorized as early or feature-level fusion, intermediate/model-level fusion, and decision/late fusion. We review them below describing their merits and demerits and propose different ways to improve multi-modality data and multi-feature fusion for SER systems.

2.4.3. Early fusion

As shown in Fig. 3(a) early fusion strategy whose resultant representation vector from either multi modalities or multi features is treated as the input data as though it was from a single modality. Though simple with no need of representation design choices, early fusion does not have the logic to differentiate between the modalities from which the resultant vector is obtained. This strategy might also not be able to identify the relationships between the combined features or modalities which further hampers the progress of explainable SER systems. They are sensitive to different sampling rates [88] and suffer from data sparsity [23]. [83], and [22] used this kind of fusion strategy. Though simple, early fusion has a number of challenges that may hinder reliable performance in lab experiments and real-time deployments. For SER, the first challenge is the dimensional difference between the text

and audio feature vectors that often lead to data sparsity and eventual selective non-robust results especially for real-time deployment. In addition, the high dimensional feature vectors that result from the early fusion strain the available resources leading to computational complexity problems [89]. Since transformer-based multi-head attention mechanisms [43] emerged, it has been used to solve enormous challenges. To solve the high dimensional problem of early fusion of text and audio modality feature vectors, the authors of [47] used cross-attention mechanism. They used transformer encoders for each modality before concatenation with speaker embeddings to avoid direct concatenation of features thereby avoiding the data sparsity problem. Sun et al. [90] also used cross-attention mechanisms to guide one modality to attend to another modality and homogeneous relationships in each model were learned using self-attention mechanisms. In terms of the fusion of heterogeneous features of the same acoustic modality, there is a need for improvement. For the few approaches we came across, the results are not evenly robust to all the emotions involved.

2.4.4. Intermediate fusion

In intermediate fusion, the feature representation vectors are first learned from separate modalities or feature branches before fusion. If they are learned by similar types of deep learning techniques it is called homogeneous learning which is other ways referred to as heterogeneous learning. Fig. 3(b) illustrates the general intermediate fusion used in SER systems. Intermediate fusion was applied in [23,47,80] among others and a commendable performance was reported for dyadic SER. Intermediate fusion is also used to solve the problems of early fusion. Ong et al. [75] used the vision transformer on the Constant-Q transform (CQT) spectrograms and mel spectrograms with short-time Fourier transform (STFT) for further feature extraction before fusion.

2.4.5. Late fusion

As illustrated in Fig. 3(c), late fusion is applied at the classification level. Representation vectors are concatenated after homogeneous or heterogeneous learning and directly input into the classifier which learns more abstract joint features before prediction. However, this kind of fusion can involve classification without further learning. On some occasions, ensemble techniques like voting, bagging, snapshot ensembles, Bayesian model averaging, and many others are used to obtain the required values according to the performance metrics used. Voting ensembles combine the predictions of the constituent branches

by taking a majority vote for the classification of emotions. Bagging involves training multiple branches as models on different data modalities and/or multi-features, typically generated through bootstrap sampling which involves random sampling with replacement. For snapshot ensembles, snapshots are saved at different stages of the training process and then ensembled. Bayesian model averaging involves combining feature or modality branches based on their posterior probabilities. Late fusion of mel spectrograms and MFCCs in a global local feature fusion model for SER in automated driving cockpits was done in [51] after the driver's speech data collection in a dynamic driving environment. A multi-feature late fusion two-branch structure was proposed for SER in an automated driving scenario that fuses the global and local heterogeneity acoustic features of the speech utterances. Though weight configuration gives good results, it was observed in [91] that scalar weight configuration for each branch in a late fusion approach results in undesirable performance. [85,92], and [82] are some of the other literature that used late fusion.

Nonetheless, early fusion and late fusion models learn less rich information since the heterogeneous and homogeneous features are learned separately. It is worth noting that early fusion and late fusion prevent models from learning intra and inter-modality interaction characteristics respectively. On the other hand, though it ensures robust performance due to the intra and inter-modality features, intermediate-level fusion is only applied at once. This means that the heterogeneous features that are not learned before the layer at which fusion takes place will be lost in addition to the high model complexity that it causes. It ends up benefiting the high-level features more than the low-level features. These issues culminate into models that often do not perform well in real-world applications especially if subjected to environmental situations with real-time ambiance, and noise. Also, cultures, accents, and languages different from those encountered during training affect the robustness and performance of models.

We opine that the exhaustive use of learned features in each modality and robust fusion strategies may provide an efficient solution for data scarcity in emotion understanding in real-world applications. We endorse the layer-based weighted fusion strategy illustrated in Fig. 3(d) for either multi-acoustic or multi-modality fusion. An explanation of the comparative efficiency of the three mentioned fusion strategies with a layer-based weighted fusion is provided in the next section. Layer-based weighted fusion strategy involves the fusion logic for local and global learned features continuously at shallow, intermediate, and higher learning stages. This is premised on the progressive learning theory where the intra and inter-modality features of modalities involved are learned through continuous layer fusion. This solves the problem of a one-time fusion strategy of either local, intermediate, or global features. We opine that this approach improves the efficiency of speech-emotion understanding of unimodal and bimodal systems, especially in limited datasets situations, and ensures avoidance of slow convergence during training. Since multi-branch weight scoring was reported to have slightly undesirable results in [91], we recommend a per layer-based weighted fusion for better results.

3. Discussion and analysis of SOTA results

In this section, we discuss the results of the different state-of-the-art approaches to SER data scarcity strategies to analyze their efficiency and robustness. We also present the challenges in the application of the discussed strategies and their possible solutions.

3.1. Comparison of results of the SOTA data scarcity strategies

A summary of the performance results of 37 published articles sampled out of the 105 articles we reviewed for SER data scarcity strategies is presented in Tables 1, 2, and 3. Table 1 presents the analysis of results of feature selection, data augmentation and domain adaptation. The choice of these papers depended on relevance in terms

of our goal and the use of common datasets for fair comparison. The summary of bimodal fusion is presented in Tables 2, and multi-acoustic feature fusion in Table 3. They are presented in terms of weighted accuracy (WA), and unweighted accuracy (UA) with details of the data scarcity strategies, and the datasets used. For better comparison, feature selection methods in this table exclude those that use attention mechanisms since they have been mostly used with fusion strategies. From the results we sampled and illustrated in the tables, it is evident that researchers have indeed utilized feature selection, data augmentation, domain adaptation, and fusion strategies to enhance performance in emotion prediction on limited SER datasets. However, it is also evident that the reported performance depends on a variety of factors that we discuss in this section. We also propose different improvements to some fusion strategies that may improve the individual emotion state prediction rate which is a general challenge to all strategies deployed.

The feature selection model proposed in [33] was evaluated on the EMODB, SAVEE, and IEMOCAP datasets registering a commendable performance of above 81.30% which shows the importance of CFS in solving small and imbalanced datasets challenges. The ELM based on a genetic algorithm for feature selection proposed in [37] achieved a minimum accuracy of 93% and a maximum of 100% in all the tested scenarios using the EMODB dataset. Besides correlation, filter, and evolution algorithms feature selection, a machine learning-based algorithm presented in [93] exhibits good results when evaluated on small datasets by selection of a high number of emotionally rich features. The authors used the ensemble random forest trees (ERFTrees) to achieve a commendable recognition accuracy of above 72.25% with nine very small datasets in Chinese language (Mandarin) hence solving the data scarcity challenges in this language for SER. They selected an acoustic feature subset of 16 out of a full set of 84 that are intensity and pitch-related. Recently, SER systems based on CNN transformers and multi-dimensional attention mechanisms were proposed in [94] for feature selection. The CNNs take care of local feature learning as the attention mechanisms learn information at different scales in the time, channel, and space dimensions. Indeed, the model's positive impact on small datasets of IEMOCAP and EMODB is shown in the performance evaluation where the model achieved 90.65% and 71.64% of unweighted accuracy on EMODB and IEMOCAP respectively. However, more work needs to be done in the reduction of the parameters of the model and improvement of the individual emotional prediction rate of the fear and neutral emotion states respectively. Nonetheless, though a very good performance is reported for feature selection strategies, they assume linearity, suffer from overfitting, introduce bias in feature importance, and may lead to loss of important information.

From the results of the literature considered, traditional and GAN-based augmentation benefits models with limited SER data, especially in terms of solving overfitting and vanishing gradient problems. However, the performance is more degraded when traditional methods are used. This is because of the dissimilarity between real and synthetic information emotion states that often increase the model confusion of emotions. Such a problem is caused by the failure to learn the data distribution by the models. The distribution challenge between synthetic and real data exists in the GAN-based augmentation too. This could be because of mode collapse [67]. Due to the data distribution challenges, the results shown in Table 1 for the SER performance on synthetically generated data using the star GAN and cycle GAN [68] need improvement. They report 76.56% as the highest accuracy for speaker multi-emotion scenarios and below 55% for multi-gender dissimilar and similar speakers. For a fair comparison with models that use other datasets, we analyze the results reported for the multi-speakers-multi-emotions experiments with gender independence. An analysis of the emotion recognition results when augmentation was applied in [60,61] shows commendable performance. An improved performance of 84.49% and 66.92% of weighted accuracy on EMODB and IEMOCAP respectively is reported in [61]. These relatively good

Table 1

Analysis of the results of the feature selection, data augmentation and Domain adaptation as SOTA SER data scarcity approaches.

Ref.	Data Scarcity Strategies	Datasets	Emotions	WA(%)	UA(%)
[33]	Feature selection (CFS) with domain adaptation	EMODB	7	–	95.10
	Feature selection (CFS) with domain adaptation	SAVEE	7	–	82.10
	Feature selection (CFS) with domain adaptation	IEMOCAP	4	–	83.80
	Feature selection (CFS) with domain adaptation	RAVDESS	8	–	81.30
[37]	Feature selection (genetic algorithms)	EMODB	7	–	93.26
[35]	Feature selection (ILFS)	EMODB	7	–	76.9
	Feature selection (Fisher)	SAVEE	7	–	42.4
	Feature selection (Fisher)	EMOVO	7	–	41.0
[30]	Feature selection (kernel PCA)	EMODB	5	–	96.0
[55]	Data augmentation (speed and volume perturbation)	IEMOCAP	4	–	76.5
	Data augmentation (speed and volume perturbation)	RAVDESS	4	–	87.3
	Data augmentation (speed and volume perturbation)	CMU-MOSEI	4	–	77.8
[21]	Data augmentation (SpecAugment, MixUp, speed perturbations)	IEMOCAP	4	–	68.1
	Data augmentation (SpecAugment, MixUp, speed perturbations)	MSP-IMPROV	4	–	62.1
[68]	Data augmentation (generative adversarial networks)	ESD	4	–	51.04
[60]	Data augmentation (generative adversarial networks)	IEMOCAP	4	66.80	62.83
[61]	Data augmentation (generative adversarial networks)	IEMOCAP	4	66.92	64.51
	Data augmentation (generative adversarial networks)	EMODB	7	84.49	83.31
[69]	Data augmentation with domain adaptation	IEMOCAP	4	–	66.06
	Data Augmentation with Domain adaptation	RAVDESS	7	–	81.29
[95]	Domain adaptation	IEMOCAP	4	–	71.7
[19]	Domain adaptation	Malayalam	4	–	88.89
[21]	Domain adaptation with auxiliary tasks	IEMOCAP	4	68.5	66.7
	Domain adaptation with auxiliary tasks	MSP-IMPROV	4	62.5	60.2

Table 2

Analysis of the results of the bimodal fusion SOTA SER data scarcity approaches.

Ref.	Data Scarcity Strategies	Datasets	Emotions	WA(%)	UA(%)
[84]	Bimodal fusion (intermediate) with attention mechanism	IEMOCAP	4	72.5	70.9
[92]	Bimodal fusion (feature and decision) with attention mechanism	IEMOCAP	4	71.06	72.05
[81]	Bimodal fusion (intermediate) with attention mechanism	IEMOCAP	4	74.33	73.23
[96]	Bimodal fusion (feature)	IEMOCAP	4	–	74.5
[97]	Bimodal fusion (intermediate)	IEMOCAP	4	–	72.6
	Bimodal Fusion (Intermediate)	MSP-Podcast	4	–	56.0
[90]	Bimodal fusion (intermediate)	IEMOCAP	7	61.2	56.0
[47]	Bimodal fusion (intermediate) with attention mechanism	IEMOCAP	4	83.6	–
	Bimodal fusion (intermediate) with attention mechanism	IEMOCAP	6	68.0	–
	Bimodal fusion (intermediate) with attention mechanism	MELD	7	62.0	–
[98]	Bimodal fusion (intermediate) with attention mechanism	IEMOCAP	7	72.8	62.5
	Bimodal fusion (intermediate) with attention mechanism	IEMOCAP	4	82.4	80.6
[99]	Bimodal Fusion (feature and decision) with attention mechanism	IEMOCAP	4	74.98	75.05
[100]	Bimodal fusion (intermediate) with attention mechanism	IEMOCAP	4	74.8	–
	Bimodal Fusion(intermediate) with attention mechanism	CMU-MOSI	2	79.85	–
	Bimodal fusion (intermediate) with attention mechanism	MELD	7	63.85	–
[23]	Bimodal Fusion (intermediate) with attention mechanism	IEMOCAP	4	75.50	75.82
[80]	Bimodal fusion (decision) with attention mechanism	IEMOCAP	4	80.51	79.22
[101]	Bimodal fusion (decision) with attention mechanism	IEMOCAP	4	73.5	73.0
[102]	Bimodal fusion (intermediate) with attention mechanism	IEMOCAP	4	76.6	76.8
[47]	Bimodal fusion (intermediate) with attention mechanism	IEMOCAP	4	83.6	83.3
	Bimodal Fusion (Intermediate) with attention mechanism	MELD	7	62.0	–
[103]	Bimodal fusion (decision) with domain adaptation	IEMOCAP	4	73.5	71.0
[104]	Bimodal fusion (decision) with domain adaptation	IEMOCAP	4	85.5	80.7
[70]	Bimodal fusion (decision) with attention mechanisms and domain adaptation	IEMOCAP	4	–	75.58
	Bimodal Fusion (decision) with attention mechanisms and domain adaptation	EMODB	7	–	84.50
[105]	Bimodal fusion (intermediate) with domain adaptation	IEMOCAP	4	72.8	–
	Bimodal Fusion (Intermediate) with domain adaptation	CMU-MOSEI	6	99.2	–
	Bimodal Fusion (Intermediate) with domain adaptation	MELD	7	63.8	–

Table 3
Analysis of the results of the multi-acoustic feature fusion SOTA SER data scarcity approaches.

Ref.	Data Scarcity Strategies	Datasets	Emotions	WA(%)	UA(%)
[22]	Acoustic feature fusion (feature) with attention mechanism	RAVDESS	4	–	85.89
	Acoustic feature fusion (feature) with attention mechanism	EMODB	4	–	95.93
	Acoustic feature fusion (feature) with attention mechanism	SAVEE	4	–	93.75
[27]	Acoustic feature fusion (intermediate) with attention mechanism	RAVDESS	7	93.65	94.03
	Acoustic feature fusion (intermediate) with attention mechanism	CREMA	6	84.00	84.00
	Acoustic feature fusion (intermediate) with attention mechanism	ASVP-ESD	6	84.40	83.40
[106]	Acoustic feature fusion (early)	RAVDESS	8	–	95.5
[54]	Acoustic feature fusion (early) and feature selection	RAVDESS	8	–	86.97
	Acoustic feature fusion (early) and feature selection	SAVEE	7	–	88.79
	Acoustic feature fusion (early) and feature selection	EMOVO	7	–	89.24
	Acoustic feature fusion (early) and feature selection	EMODB	7	–	95.29
[73]	Acoustic feature fusion (early)	IEMOCAP	4	76.54	–
	Acoustic feature fusion (early)	ESD	5	98.0	–

results are due to the use of the Wasserstein distance rather than cross-entropy loss for comparison of the real and synthetic data generated distributions. Nonetheless, data augmentation needs clear boundaries of emotions to produce better results with minimal synthesized data for better performance. An analysis of the results obtained in [107] shows that though the quality of synthetic SER data is controlled, it follows the distribution of the source data. This was evidenced in the comparison of the 80% accuracy obtained on the synthetic results in comparison with the 92.4% accuracy obtained on real data from the RAVDESS and TESS datasets.

As observed from the summarized results, data augmentation has merits in enhancing SER however, a solution to its drawbacks is paramount. Recently, Latif et al. [21] reported an average of 68.7% when SpecAugment, speed perturbations, and mixup were applied to IEMOCAP in a multi-task learning model. A similar approach was done in [61]. However, researchers have also supplemented this strategy with domain adaptation through transfer learning. Transfer learning is combined with spectrogram augmentation to re-purpose a ResNet34 model trained for speaker recognition for SER. This approach achieves an accuracy of 66.02% on the IEMOCAP dataset. Other recent methods utilize transfer learning separately. The method proposed in [95] outperforms other methods that do not use pre-trained automatic speech recognition (ASR) models to achieve an average accuracy of 71.7%. In [19], transfer learning based on TDNN for SER in low-resource languages was used with 88.89% accuracy. This points to the fact that pre-trained models provide distinguished capabilities that would otherwise not be achieved on limited datasets by the models.

Nonetheless, the performance of the models that use data augmentation, and domain adaptation alone are limited in the performance. This is evidenced in the accuracy values exhibited and the individual emotion prediction rates which affect the model's generalization capabilities. Though feature selection presents tremendous performance it is affected by the challenges of linearity assumption, and a very high computational complexity and cost caused by their operation. Therefore, there is a need for improvement in the performance. A tradeoff between performance and complexity needs to be addressed as these strategies are utilized in SER systems. Fusion strategies are a good substitute for these strategies.

Most of the recent research works have embraced fusion strategies with attention mechanisms to enhance performance under data scarcity. The sampled results shown in Tables 2 and 3 illustrate that a careful combination of the fusion strategies and attention mechanisms benefit SER in limited datasets scenarios more compared to the other strategies. A combination of domain adaptation especially transfer learning and fusion strategies have also been used to enhance SER in data scarcity. This has also been done with feature selection and data augmentation strategies. In [103], decision level fusion was used for TDNN, BiLSTM for audio, and pre-trained models for text to enhance better performance and was evaluated on IEMOCAP dataset exhibiting a recognition accuracy of 73.5%. A similar technique is reported

in [105]. The authors utilized RoBERTa and inception-ResNet-V2 with the intermediate fusion of text and audio to attain a commendable performance. The authors of [104] applied transfer learning in the text branch through the use of the BERT model along the CNN branch for the audio modality to enhance SER in limited datasets scenarios with commendable performance of 85.5% and 80.7% of unweighted and weighted accuracy. These results show that fusion strategies improve the capabilities of the other data scarcity strategies discussed earlier. More results that involve bimodal fusion strategies and attention mechanisms are summarized in Table 2.

However, most research works used a bimodal fusion of the text and audio modalities for SER compared to intra-modality heterogeneous acoustic multi-feature fusion. We recommend that acoustic multi-feature fusion should also be used by researchers not only at the feature level but also at intermediate and late fusion to avoid data or feature sparsity and loss of emotionally rich modality representations. [22,27, 108–110] are some of the few research works that have utilized this approach in acoustic feature fusion and their performance has also been summarized in Table 3. This approach benefits from the utilization of heterogeneous cues in the interaction between features of the same modality. A comparison between the performance of the models proposed in [22,27] done by us proves this assertion. A performance improvement of 8.14% was registered when both models were evaluated on RAVDESS datasets. In addition to MFCCs, mel spectrograms and chromagrams fused at the feature level in [22], the authors of [106] fused spectral contrast, and zero crossing rate (ZCR) using the deep stride CNN and fed the resultant feature representation to BiLSTM with a commendable performance. The deep stride CNN removes the need for pooling layers that usually cause loss of emotionally rich information. Though not included in Table 3 due to the difference in metrics used, acoustic graph-based fusion benefits SER. In a graph-based feature fusion proposed in [110], SER is enhanced by using each feature as a node and learning the relationship between multidimensional edge features. An improvement of about 13% is reported compared to other feature and late fusion approaches. In addition, a combination of feature selection and fusion strategies has also benefited acoustic SER with limited datasets. Commendable performance was obtained by the two-stage feature selection and early fusion strategy deployed in models proposed in [54]. Recognition accuracies of 86.97%, 88.79%, 89.24%, and 95.29% on RAVDESS, SAVEE, EMOVO, and EMOVB were reported respectively. This further affirms that a combination of data scarcity strategies does not only benefit multi-modality fusion but also acoustic feature fusion. Nonetheless, there is room for improvement, especially in terms of the individual prediction rate of negative valence emotions of fear, sadness, and disgust in the English datasets. In addition, the commendable performance is obtained with a trade-off between complexity and accuracy. Better models that utilize less computational resources with a smaller number of parameters ought to be used for SER since the models are deployed in low-resource devices.

The analysis of the results shows that fusion strategies and feature selection methods yield a good performance. However, feature selection has the disadvantage of linearity assumptions, a large number of parameters, and loss of affective information. Intermediate fusion appears to benefit most of the approaches in which it is applied [23] without the burden of computational complexity witnessed in feature selection, transfer learning, and data augmentation. This is because it ensures fast convergence and a better individual prediction rate compared to using either feature or decision-level fusion strategies. Nonetheless, SER data and feature fusion have several challenges that need to be carefully considered. These include robust representation, translation, alignment, and data sparsity. To solve these, we recommend that models should learn the homogeneous and heterogeneous relationships of emotion states in each kind of features that exist in the different modalities. In addition, the acoustic modality has heterogeneity features at the shallow, intermediate, and higher levels of the deep networks that need to be fused for better representation and eventual learning.

3.2. Challenges of data scarcity strategies and future work

Though feature selection enhances SER especially when combined with other data scarcity strategies, researchers need to be aware of its challenges which can be solved depending on the purpose of the models. Since SER involves limited datasets, feature selection may lead to overfitting. This explains why in most of the reviewed literature, feature selection is combined with either transfer learning or fusion strategies. This increases the computational complexity of the model and a tradeoff needs to be made if this strategy is chosen. In addition, feature selection results in unstable features and eventual loss of emotionally rich information. This is due to the different runs during the selection process that result in a variety of subsets of selected features. The performance of feature selection strategies especially correlation feature methods and evolutionary algorithms is also hindered by the assumption of feature independence between speech features which may fail to capture the feature relationships that may exist. This is because the emotional features have been proven to be interdependent for better results [54].

The data augmentation strategies face a variety of challenges too depending on the technique used. Traditional SER data augmentation techniques like AWGN, SpecAugment, speech perturbations, mixup may not be effective. They may not capture the variation in real and synthetic data and therefore misrepresent the emotion states which escalates model confusion. This is because some of the augmentation strategies distort or mask important emotional features that are important for emotional recognition. The GAN-based augmentation suffers from mode collapse since the real data is limited and often imbalanced [107]. There is usually a mismatch between the distribution of the real and synthetic data which yields distorted data with dissimilar emotions compared to the intended. In most cases, the bias that may exist in the real data is transferred to the synthetically generated data. In addition, speech emotions are context sensitive and the augmentation techniques may fail to generate the context and complexity that exists in the real emotional expressions which hinders the efficiency of the emotion recognition systems.

Domain mismatch between the source and target is one of the challenges of domain adaptation. In recent SER research works, the target is usually from speech recognition, speaker identification tasks, and inter-corpus and/or language SER. However, transfer learning from a domain with a different emotional state distribution may not be effective since emotions are influenced by culture, accent, and other individual differences. In situations where the pre-trained model performs worse than a model trained from scratch, there may be negative knowledge transfer. Also, fine-tuning models on a target domain with very limited labeled data may result in overfitting. These challenges are often solved by researchers through the use of a combination of other

data scarcity strategies with transfer learning. However, as observed in the discussion of the SOTA results, fusion strategies tend to perform better than the use of domain adaptation even in combination with other strategies except for feature selection.

Multi-modal data or multi-feature fusion may lead to computational complexity so strategies to avoid it are necessary. The complexity is due to the difference in modality features representations and time scales. In addition, there is always a need to take care of the alignment of all the fused modalities and features to avoid performance degradation. Data sparsity is another challenge of fusion strategies in deep learning models. This is especially due to the difference in dimensions of the modalities. Fusion models tend to rely on the modality with more data which leads to a reduction in the performance. This is also due to the imbalanced nature of the resultant feature representations of the different modalities. Some of the SOTA research works including [47,54], and [82] have handled the challenge by utilizing multi-stream intermediate fusion. In addition, proper utilization of the limited datasets through separate learning of spatial, temporal, spectral, and semantic features where possible [23] has helped to improve the performance of SER systems.

On the whole, addressing these challenges effectively requires careful balancing of data augmentation methods, robust utilization of transfer learning techniques, and ensuring that feature or data fusion methods align well with the available data and emotional cues in SER with moderate complexity that can be easily deployed in low-resource devices.

4. Conclusion

In this paper, we reviewed the data scarcity strategies employed in the state-of-the-art literature to address the challenges of limited and imbalanced datasets in SER. Specifically, we discussed the effectiveness of feature selection, data augmentation, domain adaptation, and fusion strategies, comparing their outcomes across various research works published in the last decade. While each strategy offers significant benefits for mitigating data scarcity in SER, we argue that fusion strategies, particularly those combined with attention mechanisms, are especially advantageous. These approaches promote faster convergence and reduce model complexity, thereby mitigating overfitting while performing implicit feature selection through attention mechanisms. Although intermediate fusion has consistently outperformed other fusion techniques, we recommend extending it to layer-based weight score fusion. This method would enable the model to utilize significant information from different layers, minimizing the loss of relevant features at shallow, intermediate, and deeper layers during training. In the future, a more comprehensive investigation into layer-based fusion strategies should be pursued, especially for acoustic and multimodal SER, to fully harness the potential of these approaches.

CRedit authorship contribution statement

Samuel Kakuba: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Dong Seog Han:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Conflict of interest

The authors declare that there is no conflict of interest in this paper.

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